

Website RAG Customer Support Chatbot

An n8n pipeline with two connected parts: a knowledge-base ingestion flow that crawls a website into Pinecone, and a Gemini-powered chatbot that answers only from that indexed content.



1 Executive Summary

Most support chatbots either hallucinate answers or need every FAQ typed in by hand. This workflow solves both problems: it crawls the business's own website automatically, turns the content into searchable vector embeddings, and gives a Gemini agent a strict rule — answer only from what it finds in that indexed content, and say so honestly when it can't find something. Re-running the ingestion flow keeps the chatbot's knowledge in sync with the live site.

2 Key Metrics

Ingestion Trigger	Manual trigger — run whenever the website content changes
Chat Trigger	n8n chat webhook — public, embeddable widget
Crawling	Firecrawl maps up to 100 URLs from the target site
Chunking	~1,000-character chunks with 200-character overlap, split on Unicode code points
Vector Store	Pinecone, cleared and rebuilt on every ingestion run to prevent duplicate chunks
Embeddings & LLM	Google Gemini — the same embedding model used for both indexing and retrieval
Grounding Rule	The agent must answer business questions from Pinecone only, never from memory

100

URLS CRAWLED PER RUN

10

MEMORY TURNS RETAINED

0

STALE VECTORS KEPT

3 Business Value

- ✓ **Always-current answers** — re-running the ingestion flow re-indexes the live site, so answers never come from stale FAQs.
- ✓ **No manual knowledge-base authoring** — the entire site's content becomes searchable automatically.
- ✓ **Honest about its limits** — the agent says it couldn't find something and offers a support ticket, rather than guessing.
- ✓ **Safe by construction** — the namespace is fully cleared before every re-index, so old and current content never mix.

4 Architecture Overview

Two independent flows share one Pinecone index: an ingestion pipeline run on demand, and an always-on chat flow.

FLOW	TRIGGER	PURPOSE
Knowledge-base Ingestion	Manual trigger, on demand	Crawls the website, cleans and chunks the content, embeds it, and stores it in Pinecone
Customer Support Chat	Chat webhook, always on	Answers visitor questions using only content retrieved from the indexed knowledge base

5 How It Works

Knowledge-base Ingestion	Customer Support Chat
1 Before crawling, every existing vector in the target Pinecone namespace is deleted. 2 Firecrawl crawls the root URL and returns up to 100 discovered page URLs. 3 Blocked-keyword pages, XML files and duplicate URLs are filtered out. 4 Each page is fetched, cleaned, and skipped if it reduces to under 50 characters. 5 Usable text is chunked (~1,000 chars, 200 overlap) and embedded into Pinecone with its source URL.	1 A public n8n chat trigger accepts visitor messages. 2 A Gemini agent named "Alicia" is required to use the Pinecone tool for any business question. 3 If retrieval returns nothing useful, the agent replies with one exact fallback message — never a guess. 4 A windowed memory keeps the last 10 turns so follow-up questions work naturally.

6 Reliability & Security Design

◆ **Destructive step is deliberate and scoped** — the namespace wipe only targets the configured namespace, preventing duplicate chunks.

◆ **Strict grounding, no hallucination by design** — business information must come from the retrieval tool, with an exact fallback message.

◆ **Polite, resilient crawling** — pages are processed one at a time with a rate-limit delay, and failures retry automatically.

◆ **Unicode-safe chunking** — splitting by code points prevents corrupted text from emoji or special characters.

◆ **Minimal memory footprint** — the agent stores only confirmed contact details after a booking, never company information.

◆ **Safe re-indexing** — re-running ingestion is idempotent — the namespace is always rebuilt from scratch.

7 Setup Checklist

☒ Connect credentials: Firecrawl API, Pinecone API, and Google Gemini (PaLM) API.

☒ Replace the placeholder URL in Map Website URLs with the real site to index.

☒ Create a Pinecone index whose dimension matches the chosen Gemini embedding model.

☒ Review blockedKeywords in Filter and Normalize URLs and adjust to your site's structure.

☒ Run the ingestion flow once manually and confirm vectors appear before publishing the chat trigger.

8 Node-by-Node Map

Namespace Reset	Execute Workflow → Get Pinecone Index Host → Clear Pinecone Namespace
Crawl & Filter	Map Website URLs (Firecrawl) → Filter and Normalize URLs
Fetch & Chunk	Process URLs One by One → Fetch Page Content → Clean HTML to Text → Split Text into Chunks
Chat	Chat Message Received → Customer Support Agent (Gemini + Memory + Pinecone Tool)

SUMMARY

Designed and built end-to-end in n8n, demonstrating a full RAG pipeline: automated web crawling, safe text chunking, vector indexing, and a retrieval-grounded conversational agent.